

SOUTHEAST UNIVERSITY

Department of Computer Science and Engineering

CSE 472

Web and Internet Programming Lab

Lab Report 02

HTML Fundamentals and Structured Webpage Development

ASSIGNMENT TITLE

SEU Tech Event Information
and Registration Page

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Semester	9th
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Submission date	8 September 2026

Individual laboratory submission

HTML5 structure, lists, tables, forms and semantic layout

1 Student Information

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Course	CSE 472 – Web and Internet Programming Lab
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Submission date	8 September 2026
Repository folder	https://github.com/Trisha105/CSE472/tree/main/labs/lab-02-html

2 Introduction

This experiment applies the final submission task in the CSE 472 HTML lab manual: a complete single-page event information and registration website [1]. The page is titled *SEU Tech Event Information and Registration Page*. Its content includes an event overview, goals, activity tracks, a four-session schedule, a registration form and contact details.

The event name, dates, sessions and rooms are classroom examples, not an official university announcement. The sample contact address comes from the manual and is labelled as an example. HTML provides the structure, an external CSS file provides the presentation, and a small JavaScript helper prevents the demonstration form from sending information.

3 Objectives

The aim was to build a structured HTML5 page using meaningful semantic elements and a clear heading hierarchy. The work also practised ordered, unordered and nested lists; a captioned schedule table; labelled form controls; appropriate input constraints; and correct local file paths. A further objective was to test the completed page and document its actual output with source code and browser screenshots.

4 Software and Tools Used

Tool	Use in this experiment
UTF-8 files and command-line editing	Preserving and completing the existing HTML/CSS project.
Chromium 143.0.7499.4	Rendering the website and exercising its controls.
Python 3.13.15; Playwright 1.57.0	Automated checks and genuine browser screenshots.
Beautiful Soup 4.14.3; tinycss2 1.5.1	Inspecting markup and parsing the stylesheet.
Node.js	Syntax-checking the form helper.
pdfLaTeX, listings and pdfinfo	Compiling the report, displaying full source code and checking PDF metadata.
Git and GitHub Actions	Running the actual served website, reviewing related changes and publishing the submission.

5 Short Theory of HTML

HTML stands for HyperText Markup Language. It describes the structure of webpage content and is not a general-purpose programming language. A browser reads elements such as headings, paragraphs, images, lists, tables and forms and renders them as a page. Attributes provide additional information, such as an image path, a link destination or an input type [1].

Semantic elements describe the role of content. For example, **nav** identifies navigation, **main** contains the main page content, and **footer** holds closing information. A **div** groups a block without assigning a special semantic role, while a **span** groups a short inline part of text.

6 HTML5 Document Structure

The file begins with `<!DOCTYPE html>` and uses `<html lang="en">`. Its head contains UTF-8 encoding, a responsive viewport, a descriptive browser-tab title, a page description and an external CSS link. The local SVG is also used as the favicon. The deferred helper runs after the document is parsed.

The body contains a keyboard skip link, a header with exactly one **h1**, a navigation bar, a main area with five sections and a footer. About, Goals, Activities, Schedule and Registration have IDs used by the internal links. The document therefore meets the manual's requirement for at least three meaningful sections [1].

7 Important Tags and Attributes

Tag or attribute	Use in this implementation
header, nav, main, section, footer	Separate the introduction, links, main content and contact area.
h1, h2, h3, p	Give the page a main heading, section/subsection headings and paragraphs.
div, span	Group cards and fields; highlight the purpose and event details.
a, href, target, rel	Connect page anchors and external sources; safe new-tab attributes are used.
img, src, alt	Display the local event illustration with a meaningful description.
ul, ol, li	Present goals, participation steps and nested activity tracks.
table, caption, thead, tbody	Organise the table title, column headings and four session rows.
tr, th, td, scope	Define consistent rows, header associations and data cells.
form, label, input, button	Collect demonstration values and provide Submit and Reset.
fieldset, legend	Name the participation and optional-interest groups.
id, for, name, value	Associate labels, identify fields and distinguish selected choices.
required, type, minlength, maxlength, pattern, min, max	Apply native required, length, numeric-pattern, email and date constraints.

8 Requirements Analysis

Table 1 maps the manual's final task to this project. Responsive styling, a reset button, keyboard support and safe form feedback are supporting additions, not extra requirements attributed to the manual.

Table 1: Final-task requirement mapping.

Requirement	Implemented evidence
HTML5 and semantic layout	Complete skeleton, header, nav, main, five sections and footer.
Meaningful headings and text	One h1, clear subheadings and explanatory event/participation content.
Div, span, hyperlink and image	Grouped content, highlighted text, anchors and a local SVG with alt text.
Three list types	Goals list, ordered steps and correctly nested activity tracks.
Schedule table	Caption, five column headings and four data rows.
Required registration controls	Name, ID, email, date, one radio group and three checkboxes.
Code and submission evidence	Full source, five browser screenshots, compiled report and verified ZIP.

9 Project Folder Structure

Listing 1: Final Lab 02 folder layout.

```
labs/lab-02-html/  
|-- index.html  
|-- style.css  
|-- form-demo.js  
|-- images/event-banner.svg  
|-- screenshots/  
|   |-- homepage-top.png  
|   |-- event-goals.png  
|   |-- event-details.png  
|   |-- registration-form.png  
|   '-- mobile-registration.png  
|-- report/  
|   |-- lab_report_02.tex  
|   |-- build-metadata.tex  
|   |-- test-cases.tex  
|   '-- lab_report_02.pdf  
|-- tests/  
|   |-- results.json  
|   '-- TEST_RESULTS.md  
|-- build_submission.py  
|-- README.md  
|-- SHA256SUMS.txt  
'-- CSE472_Lab_02_Complete.zip
```

The archive excludes itself, Git metadata, caches, temporary LaTeX files and unrelated repository content. SVG is used instead of the suggested JPG because it is a small scalable local image; the manual does not require a particular image format.

10 Design and Implementation Procedure

The repository was inspected before editing. The existing Lab 02 HTML, CSS and README on **main** were read. No other lab folder or repository-specific instruction file was present in the inspected baseline. The update retained the existing event content, section order, cards and navy/teal colour scheme.

The missing image was supplied as an original SVG, and its reference was corrected. A skip link, labelled scrollable table region, reduced-motion rule and local form feedback were added. The ID field now accepts 10–15 digits; this is an example validation rule, not a claim about university-wide ID policy.

The helper installs submission/reset handlers before enabling Submit. Native HTML validation runs first; a valid submission is then cancelled and replaced by a local demonstration message. The completed page was tested, captured in Chromium, included in this report and packaged. Only related Lab 02 files and its build workflow were intentionally changed.

11 Complete HTML Source Code

Listing 2 contains all 282 lines of the final HTML file. Visual line wrapping in this report does not change the source.

Listing 2: Complete final index.html.

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <meta name="description" content="SEU Tech Event information, schedule, activities, and student
7     registration
8     form for CSE 472 Lab 02.">
9   <title>SEU Tech Event Information and Registration Page</title>
10  <link rel="stylesheet" href="style.css">
11  <link rel="icon" href="images/event-banner.svg" type="image/svg+xml">
12  <script src="form-demo.js" defer></script>
13 </head>
14 <body>
15   <a class="skip-link" href="#main-content">Skip to main content</a>
16   <header class="site-header" id="top">
17     <div class="header-inner">
18       <p class="eyebrow">CSE 472 | Web and Internet Programming Lab</p>
19       <h1>SEU Tech Event Information and Registration Page</h1>
20       <p class="lead">A student-focused technology showcase featuring practical sessions, project
21         demonstrations, and collaborative learning at Southeast University.</p>
22       <div class="hero-meta" aria-label="Event summary">
23         <span>Sample event: 15-16 October 2026</span>
24         <span>SEU CSE Department</span>
25         <span>Open to SEU students</span>
26       </div>
27     </div>
28   </header>
29   <nav class="main-nav" aria-label="Primary navigation">
30     <div class="nav-inner">
31       <a href="#about">About</a>
32       <a href="#goals">Goals</a>
33       <a href="#activities">Activities</a>
34       <a href="#schedule">Schedule</a>
35       <a href="#register">Register</a>
36       <a href="#contact">Contact</a>
37     </div>
38   </nav>
39   <main id="main-content" class="page-shell" tabindex="-1">
40     <section id="about" class="section-card about-grid">
41       <div>
42         <p class="section-kicker">Event overview</p>
43         <h2>About the Event</h2>
44         <p>SEU TechConnect 2026 is a demonstration event created for this HTML lab assignment. The page
45           shows
46           how a university technology event can present useful information in a clear and structured
47           format.</p>
48         <p>The <span class="text-highlight">main purpose</span> is to bring students together around
49           web
50           development, artificial intelligence, robotics, and practical problem solving. Participants
51           can
52           attend sessions, view student projects, and register their area of interest.</p>
53         <div class="info-box">
54           <h3>Participation Details</h3>
55           <p><strong>Eligibility:</strong> Current Southeast University students from any department.</p>
56           <p><strong>Venue:</strong> Department of Computer Science and Engineering, Southeast
57             University.</p>
58           <p><strong>Participation:</strong> Registration is free for this classroom demonstration page.
59           </p>

```

```

56     <p><a class="external-link" href="https://www.seu.edu.bd" target="_blank" rel="noopener
57         norereferrer">Visit the Southeast University website</a></p>
58 </div>
59 </div>
60 <figure class="event-figure">
61     
63     <figcaption>Original event banner prepared for this Lab 02 webpage.</figcaption>
64 </figure>
65 </section>
66
67 <section id="goals" class="section-card">
68     <p class="section-kicker">Why the event matters</p>
69     <h2>Event Goals</h2>
70     <div class="two-column">
71         <div>
72             <h3>Core Goals</h3>
73             <ul class="feature-list">
74                 <li>Encourage students to share small technology projects and practical ideas.</li>
75                 <li>Introduce current areas such as web development, AI, data analysis, IoT, and robotics.<
/li>
76                 <li>Build confidence through short presentations, teamwork, and peer discussion.</li>
77                 <li>Create a friendly space for students from different semesters to learn together.</li>
78             </ul>
79         </div>
80         <div class="mini-card">
81             <h3>What a Participant Can Gain</h3>
82             <p>Students can observe project demonstrations, discuss tools with peers, and identify topics
they
83                 may want to explore in later courses or personal projects.</p>
84             <p class="note">This page is an academic lab project and not an official event announcement.<
/p>
85         </div>
86     </div>
87 </section>
88
89 <section id="activities" class="section-card">
90     <p class="section-kicker">Program and participation</p>
91     <h2>Topics and Activities</h2>
92     <div class="two-column">
93         <div>
94             <h3>How to Participate</h3>
95             <ol class="numbered-list">
96                 <li>Read the event details and schedule.</li>
97                 <li>Select the activity track that matches your interest.</li>
98                 <li>Try the demonstration form using sample information.</li>
99                 <li>Check the validation message; no real booking is made.</li>
100             </ol>
101         </div>
102         <div>
103             <h3>Activity Tracks</h3>
104             <ul class="nested-list">
105                 <li>Software and Web
106                     <ul>
107                         <li>HTML and CSS mini showcase</li>
108                         <li>Beginner web accessibility tips</li>
109                     </ul>
110                 </li>
111                 <li>AI and Data
112                     <ul>
113                         <li>Student machine learning demos</li>
114                         <li>Data visualization discussion</li>
115                     </ul>
116                 </li>
117                 <li>IoT and Robotics
118                     <ul>
119                         <li>Sensor-based prototypes</li>
120                         <li>Line follower and automation ideas</li>
121                     </ul>

```

```

122         </li>
123     </ul>
124 </div>
125 </div>
126 </section>
127
128 <section id="schedule" class="section-card">
129     <p class="section-kicker">Planned sessions</p>
130     <h2>Event Schedule</h2>
131     <p>The following schedule is a sample program prepared for the assignment. It demonstrates a
132     structured
133     table with clear headings and multiple data rows.</p>
134     <div class="table-wrap" tabindex="0" role="region" aria-label="Scrollable event schedule">
135         <table>
136             <caption>SEU TechConnect 2026 - Sample Event Schedule</caption>
137             <thead>
138                 <tr>
139                     <th scope="col">Day</th>
140                     <th scope="col">Time</th>
141                     <th scope="col">Session</th>
142                     <th scope="col">Speaker</th>
143                     <th scope="col">Venue</th>
144                 </tr>
145             </thead>
146             <tbody>
147                 <tr>
148                     <td>15 Oct</td>
149                     <td>10:00 AM</td>
150                     <td>Opening and Web Showcase</td>
151                     <td>CSE Student Team</td>
152                     <td>Lab 1</td>
153                 </tr>
154                 <tr>
155                     <td>15 Oct</td>
156                     <td>1:30 PM</td>
157                     <td>AI and Data Demonstrations</td>
158                     <td>Student Presenters</td>
159                     <td>Lab 2</td>
160                 </tr>
161                 <tr>
162                     <td>16 Oct</td>
163                     <td>10:30 AM</td>
164                     <td>IoT and Robotics Session</td>
165                     <td>Project Groups</td>
166                     <td>Robotics Corner</td>
167                 </tr>
168                 <tr>
169                     <td>16 Oct</td>
170                     <td>2:00 PM</td>
171                     <td>Closing Discussion and Feedback</td>
172                     <td>Event Coordinators</td>
173                     <td>Seminar Room</td>
174                 </tr>
175             </tbody>
176         </table>
177     </div>
178 </section>
179
180 <section id="register" class="section-card registration-section">
181     <p class="section-kicker">Student registration</p>
182     <h2>Registration Form</h2>
183     <p>Complete the fields below to demonstrate HTML form controls and validation. This is a static
184     lab
185     form, so no information is sent to a server.</p>
186
187     <form id="registration-form" action="#register" method="get"
188         aria-describedby="static-notice" autocomplete="off">
189         <div class="form-grid">
190             <div class="field-group">
191                 <label for="fullname">Full Name</label>

```



```

190     <input type="text" id="fullname" name="fullname" placeholder="Enter your full name"
minlength="3"
191         maxlength="60" required>
192     </div>
193
194     <div class="field-group">
195         <label for="studentid">Student ID</label>
196         <input type="text" id="studentid" name="studentid" placeholder="Enter your student ID"
197             minlength="10" maxlength="15" inputmode="numeric"
198             pattern="[0-9]{10,15}" title="Enter 10 to 15 digits." required>
199     </div>
200
201     <div class="field-group">
202         <label for="email">Email Address</label>
203         <input type="email" id="email" name="email" placeholder="name@example.com" maxlength="80"
204             required>
205     </div>
206
207     <div class="field-group">
208         <label for="preferred-date">Preferred Date</label>
209         <input type="date" id="preferred-date" name="preferred_date" min="2026-10-15" max="
2026-10-16"
210             required>
211     </div>
212 </div>
213
214 <fieldset>
215     <legend>Preferred Participation Type</legend>
216     <div class="choice-row">
217         <div class="choice-item">
218             <input type="radio" id="attendee" name="participation_type" value="Attendee" required>
219             <label for="attendee">General Attendee</label>
220         </div>
221         <div class="choice-item">
222             <input type="radio" id="presenter" name="participation_type" value="Presenter">
223             <label for="presenter">Project Presenter</label>
224         </div>
225     </div>
226 </fieldset>
227
228 <fieldset>
229     <legend>Topics of Interest (optional)</legend>
230     <div class="choice-row">
231         <div class="choice-item">
232             <input type="checkbox" id="topic-web" name="topics" value="Web Development">
233             <label for="topic-web">Web Development</label>
234         </div>
235         <div class="choice-item">
236             <input type="checkbox" id="topic-ai" name="topics" value="AI and Data">
237             <label for="topic-ai">AI and Data</label>
238         </div>
239         <div class="choice-item">
240             <input type="checkbox" id="topic-iot" name="topics" value="IoT and Robotics">
241             <label for="topic-iot">IoT and Robotics</label>
242         </div>
243     </div>
244 </fieldset>
245
246     <div id="static-notice" class="form-note" role="note">
247         <strong>Static form notice:</strong> Submit checks your entries locally. No registration is
created,
248         and this page does not send or save your information.
249     </div>
250
251     <div class="button-row">
252         <button type="submit" id="submit-demo" class="primary-button" disabled>Submit Registration</
button>
253         <button type="reset" class="secondary-button">Reset Form</button>
254     </div>
255     <p id="form-status" class="form-status" role="status" aria-live="polite">Use sample information

```

```
256         to
257         test this demonstration.</p>
258     <noscript><p>JavaScript is disabled. Submit is unavailable to prevent data transmission. Reset
259     still
260     works.</p></noscript>
261 </form>
262 </section>
263 </main>
264
265 <footer id="contact" class="site-footer">
266     <div class="footer-inner">
267         <div>
268             <h2>Contact Information</h2>
269             <p>Department of Computer Science and Engineering<br>Southeast University, Dhaka, Bangladesh</p>
270         </div>
271         <p>Sample contact from the lab manual: <span>cse.lab@seu.edu.bd</span>
272         <br>This address is shown as an example; event enquiries are not monitored here.</p>
273     </div>
274     <div class="footer-note">
275         <p><strong>CSE 472 Lab Report 02</strong></p>
276         <p>Prepared by Jannatul Tuna Trisha</p>
277         <p>Student ID: 2023200000664 | Section: 01 | 9th Semester</p>
278         <p><a href="https://html.spec.whatwg.org/multipage/" target="_blank"
279             rel="noopener noreferrer">Read the HTML Living Standard</a></p>
280         <p><a href="#top">Back to top</a></p>
281     </div>
282 </div>
283 </footer>
284 </body>
285 </html>
```

12 Complete CSS Source Code

Listing 3 contains all 540 lines of the external stylesheet. The original theme is retained with small accessibility and form-state additions.

Listing 3: Complete final style.css.

```

1  :root {
2    --navy: #0f2a43;
3    --blue: #155e9a;
4    --teal: #0b7476;
5    --sky: #eaf4fb;
6    --mint: #e8f7f4;
7    --ink: #1d2a36;
8    --muted: #5e6b77;
9    --line: #d6e0e8;
10   --white: #ffffff;
11   --shadow: 0 14px 34px rgba(15, 42, 67, 0.10);
12   --radius: 18px;
13 }
14
15 * {
16   box-sizing: border-box;
17 }
18
19 html {
20   scroll-behavior: smooth;
21 }
22
23 body {
24   margin: 0;
25   font-family: "Segoe UI", Arial, sans-serif;
26   color: var(--ink);
27   background: #f5f8fb;
28   line-height: 1.65;
29 }
30
31 img {
32   max-width: 100%;
33   height: auto;
34   display: block;
35 }
36
37 a {
38   color: var(--blue);
39 }
40
41 a:hover {
42   color: var(--teal);
43 }
44
45 a:focus-visible,
46 button:focus-visible,
47 input:focus-visible,
48 .table-wrap:focus-visible {
49   outline: 3px solid #ffbf47;
50   outline-offset: 3px;
51 }
52
53 .site-header {
54   background: linear-gradient(125deg, #0f2a43 0%, #155e9a 58%, #0f8b8d 100%);
55   color: var(--white);
56   padding: 64px 24px 54px;
57 }
58
59 .header-inner,
60 .nav-inner,
61 .page-shell,
62 .footer-inner {

```

```
63   width: min(1120px, 100%);
64   margin: 0 auto;
65 }
66
67 .eyebrow,
68 .section-kicker {
69   margin: 0 0 8px;
70   font-size: 0.82rem;
71   font-weight: 700;
72   letter-spacing: 0.12em;
73   text-transform: uppercase;
74 }
75
76 .eyebrow {
77   color: #d6eef8;
78 }
79
80 .site-header h1 {
81   max-width: 920px;
82   margin: 0;
83   font-size: clamp(2rem, 5vw, 3.6rem);
84   line-height: 1.08;
85   letter-spacing: -0.02em;
86 }
87
88 .lead {
89   max-width: 820px;
90   margin: 20px 0 0;
91   font-size: 1.12rem;
92   color: #eaf7fb;
93 }
94
95 .hero-meta {
96   display: flex;
97   flex-wrap: wrap;
98   gap: 10px;
99   margin-top: 26px;
100 }
101
102 .hero-meta span {
103   padding: 8px 12px;
104   border: 1px solid rgba(255, 255, 255, 0.35);
105   border-radius: 999px;
106   background: rgba(255, 255, 255, 0.10);
107   font-size: 0.92rem;
108 }
109
110 .main-nav {
111   position: sticky;
112   top: 0;
113   z-index: 20;
114   background: rgba(255, 255, 255, 0.96);
115   border-bottom: 1px solid var(--line);
116   backdrop-filter: blur(10px);
117 }
118
119 .nav-inner {
120   display: flex;
121   flex-wrap: wrap;
122   gap: 6px 12px;
123   padding: 12px 24px;
124 }
125
126 .main-nav a {
127   padding: 7px 10px;
128   border-radius: 8px;
129   color: var(--navy);
130   font-weight: 700;
131   text-decoration: none;
132 }
```

```
133
134 .main-nav a:hover {
135     background: var(--sky);
136 }
137
138 .page-shell {
139     padding: 34px 24px 58px;
140 }
141
142 .section-card {
143     margin-bottom: 28px;
144     padding: 34px;
145     border: 1px solid #e2e9ef;
146     border-radius: var(--radius);
147     background: var(--white);
148     box-shadow: var(--shadow);
149     scroll-margin-top: 84px;
150 }
151
152 .section-kicker {
153     color: var(--teal);
154 }
155
156 .section-card h2 {
157     margin: 0 0 14px;
158     color: var(--navy);
159     font-size: clamp(1.55rem, 3vw, 2.05rem);
160 }
161
162 .section-card h3 {
163     margin-top: 0;
164     color: #174b73;
165 }
166
167 .about-grid {
168     display: grid;
169     grid-template-columns: 1.12fr 0.88fr;
170     gap: 30px;
171     align-items: start;
172 }
173
174 .event-figure {
175     margin: 0;
176     overflow: hidden;
177     border-radius: 16px;
178     border: 1px solid var(--line);
179     background: #f8fbfd;
180 }
181
182 .event-figure img {
183     width: 100%;
184     aspect-ratio: 16 / 10;
185     object-fit: cover;
186 }
187
188 .event-figure figcaption {
189     padding: 10px 14px 12px;
190     color: var(--muted);
191     font-size: 0.9rem;
192 }
193
194 .text-highlight {
195     color: var(--teal);
196     font-weight: 700;
197 }
198
199 .info-box,
200 .mini-card,
201 .form-note {
202     border-left: 4px solid var(--teal);
```

```
203     background: var(--mint);
204     border-radius: 12px;
205 }
206
207 .info-box {
208     margin-top: 22px;
209     padding: 20px;
210 }
211
212 .info-box p:last-child {
213     margin-bottom: 0;
214 }
215
216 .external-link {
217     font-weight: 700;
218 }
219
220 .two-column {
221     display: grid;
222     grid-template-columns: repeat(2, minmax(0, 1fr));
223     gap: 26px;
224 }
225
226 .mini-card {
227     padding: 22px;
228 }
229
230 .note {
231     margin-bottom: 0;
232     color: #365664;
233     font-size: 0.94rem;
234 }
235
236 .feature-list,
237 .numbered-list,
238 .nested-list {
239     padding-left: 1.35rem;
240 }
241
242 .feature-list li,
243 .numbered-list li,
244 .nested-list li {
245     margin-bottom: 8px;
246 }
247
248 .nested-list ul {
249     margin-top: 6px;
250     padding-left: 1.2rem;
251     color: var(--muted);
252 }
253
254 .table-wrap {
255     width: 100%;
256     margin-top: 18px;
257     overflow-x: auto;
258     border: 1px solid var(--line);
259     border-radius: 14px;
260 }
261
262 table {
263     width: 100%;
264     min-width: 760px;
265     border-collapse: collapse;
266     background: var(--white);
267 }
268
269 caption {
270     padding: 15px 18px;
271     text-align: left;
272     font-weight: 800;
```

```
273     color: var(--navy);
274     background: #f2f7fb;
275 }
276
277 th,
278 td {
279     padding: 14px 16px;
280     border-top: 1px solid var(--line);
281     text-align: left;
282     vertical-align: middle;
283 }
284
285 th {
286     color: var(--white);
287     background: var(--blue);
288     font-size: 0.93rem;
289 }
290
291 tbody tr:nth-child(even) {
292     background: #f8fbfd;
293 }
294
295 .registration-section form {
296     margin-top: 24px;
297 }
298
299 .form-grid {
300     display: grid;
301     grid-template-columns: repeat(2, minmax(0, 1fr));
302     gap: 18px;
303 }
304
305 .field-group {
306     display: flex;
307     flex-direction: column;
308     gap: 7px;
309 }
310
311 label,
312 legend {
313     font-weight: 700;
314     color: var(--navy);
315 }
316
317 input[type="text"],
318 input[type="email"],
319 input[type="date"] {
320     width: 100%;
321     min-height: 46px;
322     padding: 10px 12px;
323     border: 1px solid #b9c9d6;
324     border-radius: 10px;
325     background: var(--white);
326     color: var(--ink);
327     font: inherit;
328 }
329
330 input[type="text"]:invalid:not(:placeholder-shown),
331 input[type="email"]:invalid:not(:placeholder-shown) {
332     border-color: #a43b3b;
333 }
334
335 fieldset {
336     margin: 22px 0 0;
337     padding: 18px;
338     border: 1px solid var(--line);
339     border-radius: 12px;
340 }
341
342 legend {
```

```
343     padding: 0 8px;
344 }
345
346 .choice-row {
347     display: flex;
348     flex-wrap: wrap;
349     gap: 14px 22px;
350 }
351
352 .choice-item {
353     display: flex;
354     align-items: center;
355     gap: 8px;
356 }
357
358 .choice-item input {
359     width: 18px;
360     height: 18px;
361     accent-color: var(--teal);
362 }
363
364 .form-note {
365     margin-top: 22px;
366     padding: 15px 18px;
367     color: #294c57;
368 }
369
370 .button-row {
371     display: flex;
372     flex-wrap: wrap;
373     gap: 12px;
374     margin-top: 22px;
375 }
376
377 button {
378     min-height: 44px;
379     padding: 10px 18px;
380     border: 0;
381     border-radius: 10px;
382     cursor: pointer;
383     font: inherit;
384     font-weight: 800;
385 }
386
387 .primary-button {
388     color: var(--white);
389     background: var(--blue);
390 }
391
392 .primary-button:hover {
393     background: #0f4f83;
394 }
395
396 .secondary-button {
397     color: var(--navy);
398     background: #e7eef4;
399 }
400
401 .secondary-button:hover {
402     background: #d8e4ed;
403 }
404
405 .site-footer {
406     background: var(--navy);
407     color: #e8f0f5;
408     padding: 36px 24px;
409 }
410
411 .footer-inner {
412     display: grid;
```



```
413     grid-template-columns: 1.1fr 0.9fr;
414     gap: 28px;
415 }
416
417 .site-footer h2 {
418     margin-top: 0;
419     color: var(--white);
420 }
421
422 .site-footer a {
423     color: #8fe3dd;
424 }
425
426 .footer-note {
427     padding: 18px;
428     border: 1px solid rgba(255, 255, 255, 0.18);
429     border-radius: 14px;
430     background: rgba(255, 255, 255, 0.06);
431 }
432
433 .footer-note p {
434     margin-top: 0;
435 }
436
437 @media (max-width: 820px) {
438     .about-grid,
439     .two-column,
440     .form-grid,
441     .footer-inner {
442         grid-template-columns: 1fr;
443     }
444
445     .site-header {
446         padding-top: 48px;
447     }
448
449     .section-card {
450         padding: 26px;
451     }
452
453     .nav-inner {
454         overflow-x: auto;
455         flex-wrap: nowrap;
456     }
457
458     .main-nav a {
459         white-space: nowrap;
460     }
461 }
462
463 @media (max-width: 520px) {
464     .site-header {
465         padding: 40px 18px 38px;
466     }
467
468     .page-shell,
469     .nav-inner {
470         padding-left: 16px;
471         padding-right: 16px;
472     }
473
474     .section-card {
475         padding: 22px 18px;
476         border-radius: 14px;
477     }
478
479     .hero-meta {
480         display: grid;
481     }
482 }
```

```
483 .choice-row,
484 .button-row {
485     flex-direction: column;
486     align-items: stretch;
487 }
488
489 button {
490     width: 100%;
491 }
492 }
493
494 /* Small accessibility and safe-demo additions to the existing Lab 02 theme. */
495 .skip-link {
496     position: absolute;
497     top: 10px;
498     left: 10px;
499     z-index: 100;
500     padding: 10px 16px;
501     background: var(--white);
502     transform: translateY(-180%);
503 }
504
505 .skip-link:focus {
506     transform: translateY(0);
507 }
508
509 #main-content,
510 #contact {
511     scroll-margin-top: 84px;
512 }
513
514 .form-status {
515     min-height: 1.65em;
516     margin: 18px 0 0;
517     color: #205c4d;
518     font-weight: 600;
519 }
520
521 button:disabled {
522     opacity: 0.6;
523     cursor: not-allowed;
524 }
525
526 .about-grid > div,
527 .footer-inner > div,
528 .field-group {
529     min-width: 0;
530 }
531
532 .site-footer p {
533     overflow-wrap: anywhere;
534 }
535
536 @media (prefers-reduced-motion: reduce) {
537     html {
538         scroll-behavior: auto;
539     }
540 }
```

13 Explanation of Major Website Sections

Header and navigation. The header introduces the page and sample event. Six main links reach the sections and footer; a skip link and Back to top link provide additional navigation. The single main heading names the assignment.

About and Goals. About explains eligibility, venue and participation. The local illustration depicts a laptop and technology topics, not an actual campus photograph. Goals are shown as an unordered list, and the academic-project notice makes the fictional event status clear.

Activities and Schedule. An ordered list explains the demonstration process. Nested lists group Web, AI/Data and IoT/Robotics topics. The schedule has a caption, five column headings and four sample sessions. Its labelled region scrolls on narrow screens without widening the whole page.

Registration and Contact. Nine inputs are explicitly labelled. The four main fields use text, email and date types; radio buttons share one group name, while checkboxes allow multiple interests. The footer identifies the student and labels the manual's contact address as an example.

13.1 Complete Safety Helper

The helper adds no database, account system, server request or storage. Submit starts disabled, is enabled only after the handlers are installed, and remains unavailable without JavaScript. Native HTML controls supply the actual field validation [2].

Listing 4: Complete form-demo.js helper.

```

1 "use strict";
2
3 // Safety helper only: field validation remains native HTML validation.
4 // Install the handlers before enabling Submit. No storage or network calls.
5 const form = document.getElementById("registration-form");
6 const submitButton = document.getElementById("submit-demo");
7 const statusMessage = document.getElementById("form-status");
8
9 if (form && submitButton && statusMessage) {
10   form.addEventListener("submit", function (event) {
11     event.preventDefault();
12     statusMessage.textContent =
13       "Validation successful. Demo only: no information was sent or saved.";
14   });
15
16   form.addEventListener("reset", function () {
17     statusMessage.textContent =
18       "Form reset. Use sample information to try the demonstration again.";
19   });
20
21   // Without this helper, the disabled button prevents click/Enter submission.
22   submitButton.disabled = false;
23 }

```

14 Testing Procedure

Static checks inspected explicit tag nesting, metadata, landmarks, IDs, labels, list structure, schedule rows and local file references. tinycss2 parsed the stylesheet, Chromium checked its declaration support, and Node checked the helper syntax. These are local parser/browser checks, not an official W3C conformance certificate.

The final recorded run used Chromium 143.0.7499.4 with the delivered files served by a real local HTTP server in GitHub Actions. HTML, CSS, JavaScript and the SVG were loaded through their

actual URLs. The earlier local development browser had blocked direct file and loopback access; the final served-page run is a separate check and is reported independently.

Browser tests exercised required and malformed fields, date boundaries, radio exclusivity, independent checkboxes, Submit, Enter, Reset, labels, focus outlines, reduced motion and widths of 1200, 820, 390 and 320 pixels. Fictitious values were used: Demo Student, 2023000000001 and demo@example.com. The test observed the URL and network activity around submission. A separate JavaScript-disabled context checked the fallback.

External-link safety and a live visit to the HTML standard were checked separately. Any unavailable live destination is marked BLOCKED, not PASS. The university server's availability, other browser engines, physical phones and full screen-reader behaviour are not asserted. To repeat the build, install the tools in the README and run `python build_submission.py` from the Lab 02 folder.

15 Test Cases and Actual Results

The final recorded run contains 33 checks: **33 passed, 0 blocked, and 0 failed**. The following results are generated from the executed checks. Full records are included in `tests/results.json` and `tests/TEST_RESULTS.md`.

Table 2: Executed tests with expected and actual results.

ID	Check	Expected	Actual result
T01	HTML5 structure and nesting	HTML5, metadata and balanced tags	PASS. Unclosed tags: 0; nesting errors: 0
T02	Semantic structure	One h1; at least three sections; all landmarks	PASS. 1 main heading; 5 sections
T03	IDs and labels	Unique IDs and connected labels for every input	PASS. 21 IDs; 9 explicitly labelled inputs
T04	Lists	Ordered, unordered and correctly nested lists	PASS. 3 nested lists inside parent list items
T05	Schedule table	Caption, five headings and four consistent data rows	PASS. 5 headings; 4 data rows
T06	Local file references	All linked local resources exist	PASS. 4 references; missing: []
T07	CSS syntax	No CSS parser errors	PASS. 241 declarations; 0 errors
T08	Helper syntax	Node syntax check succeeds	PASS. Exit code 0
T09	Rendered image and styling	Image loaded and CSS declarations supported	PASS. Image loaded: True; unsupported declarations: 0
T10	Internal navigation	Every link reaches its target	PASS. 8 links checked; 0 failures
T11	External link attributes	HTTPS, new tab and safe relationship attributes	PASS. 2 external links checked
T12	Live external destination	HTML standard loads successfully	PASS. HTTP 200
T13	Required fields	Empty form rejected and first required field focused	PASS. Valid: False; focus: fullname
T14	Name minimum length	Two-character name rejected	PASS. tooShort: True
T15	Student ID limits	Letters rejected; maximum fifteen digits	PASS. Letter mismatch: True; capped length: 15
T16	Email validation	Malformed email rejected	PASS. typeMismatch: True
T17	Date boundaries	Only 15 and 16 October accepted	PASS. 14, 15, 16, 17 October valid: [False, True, True, False]
T18	Radio group	Exactly one participation type selected	PASS. Exclusive: True
T19	Checkbox choices	Two topics can be selected independently	PASS. Selected topics: 2
T20	Valid demo submission	Local confirmation; no request or navigation	PASS. Confirmation: True; new requests: 0
T21	Enter-key submission	Enter uses the same safe handler	PASS. Unchanged URL: True; new requests: 0

Table 2 continued

ID	Check	Expected	Actual result
T22	Reset button	All fields and selections return to defaults	PASS. All controls reset: True
T23	Label interaction	Labels focus or toggle the corresponding input	PASS. Text focus: True; checkbox selected: True
T24	Keyboard focus	Tab gives a visible three-pixel focus outline	PASS. 3px solid
T25	Responsive layout at 1200px	No horizontal document overflow	PASS. Document width: 1200px; viewport: 1200px
T26	Responsive layout at 820px	No horizontal document overflow	PASS. Document width: 820px; viewport: 820px
T27	Responsive layout at 390px	No horizontal document overflow	PASS. Document width: 390px; viewport: 390px
T28	Responsive layout at 320px	No horizontal document overflow	PASS. Document width: 320px; viewport: 320px
T29	Mobile table scrolling	Wide table scrolls inside its own region	PASS. Horizontal offset: 100px
T30	Reduced motion	Reduced-motion preference disables smooth scrolling	PASS. scroll-behavior: auto
T31	No-JavaScript safety	Submit disabled; Enter sends no information	PASS. Disabled: True; new requests: 0; unchanged URL: True
T32	Console and page errors	No avoidable errors on the implemented page	PASS. Errors recorded: 0
T33	Real resource loading	HTML, CSS, helper and SVG load through HTTP	PASS. All four resources returned HTTP 200: True

16 Output Screenshots

Figures 1–5 are genuine Chromium captures of the implemented page. Four images show desktop page regions at a 1200-pixel width; the final image shows a 390-by-844-pixel mobile viewport. No screenshot is copied from the manual or recreated as a mockup.

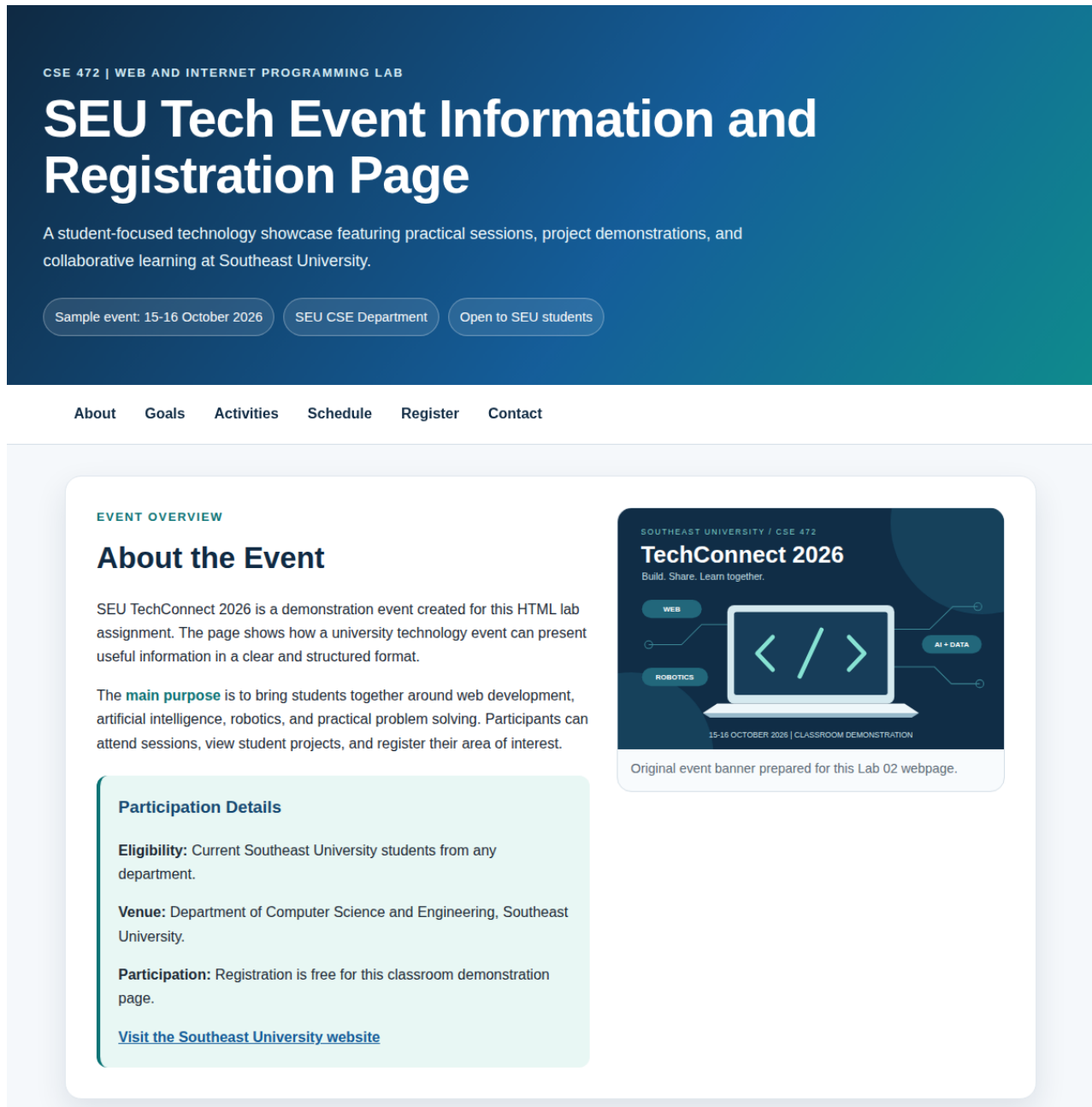


Figure 1: Header, navigation, introduction, local event illustration and participation details.

PROGRAM AND PARTICIPATION

Topics and Activities

How to Participate

1. Read the event details and schedule.
2. Select the activity track that matches your interest.
3. Try the demonstration form using sample information.
4. Check the validation message; no real booking is made.

Activity Tracks

- Software and Web
 - HTML and CSS mini showcase
 - Beginner web accessibility tips
- AI and Data
 - Student machine learning demos
 - Data visualization discussion
- IoT and Robotics
 - Sensor-based prototypes
 - Line follower and automation ideas

PLANNED SESSIONS

Event Schedule

The following schedule is a sample program prepared for the assignment. It demonstrates a structured table with clear headings and multiple data rows.

SEU TechConnect 2026 - Sample Event Schedule				
Day	Time	Session	Speaker	Venue
15 Oct	10:00 AM	Opening and Web Showcase	CSE Student Team	Lab 1
15 Oct	1:30 PM	AI and Data Demonstrations	Student Presenters	Lab 2
16 Oct	10:30 AM	IoT and Robotics Session	Project Groups	Robotics Corner
16 Oct	2:00 PM	Closing Discussion and Feedback	Event Coordinators	Seminar Room

Figure 2: Ordered participation steps, nested topic lists and the complete four-row schedule.

The activity lists are nested inside their parent items. All five headings and all four sample sessions are visible in the desktop table.

STUDENT REGISTRATION

Registration Form

Complete the fields below to demonstrate HTML form controls and validation. This is a static lab form, so no information is sent to a server.

Full Name

Demo Student

Student ID

2023000000001

Email Address

demo@example.com

Preferred Date

10/15/2026

Preferred Participation Type

☒ General Attendee ☐ Project Presenter

Topics of Interest (optional)

☒ Web Development ☒ AI and Data ☐ IoT and Robotics

Static form notice: Submit checks your entries locally. No registration is created, and this page does not send or save your information.

Submit Registration

Reset Form

Validation successful. Demo only: no information was sent or saved.

Contact Information

Department of Computer Science and Engineering
Southeast University, Dhaka, Bangladesh

Sample contact from the lab manual: cse.lab@seu.edu.bd
This address is shown as an example; event enquiries are not monitored here.

CSE 472 Lab Report 02

Prepared by Jannatul Tuna Trisha

Student ID: 2023200000664 | Section: 01 | 9th Semester

[Read the HTML Living Standard](#)

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Figure 3: Complete form and footer after a valid demonstration submission. The fictitious test values are shown with explicit confirmation that nothing was sent or saved.

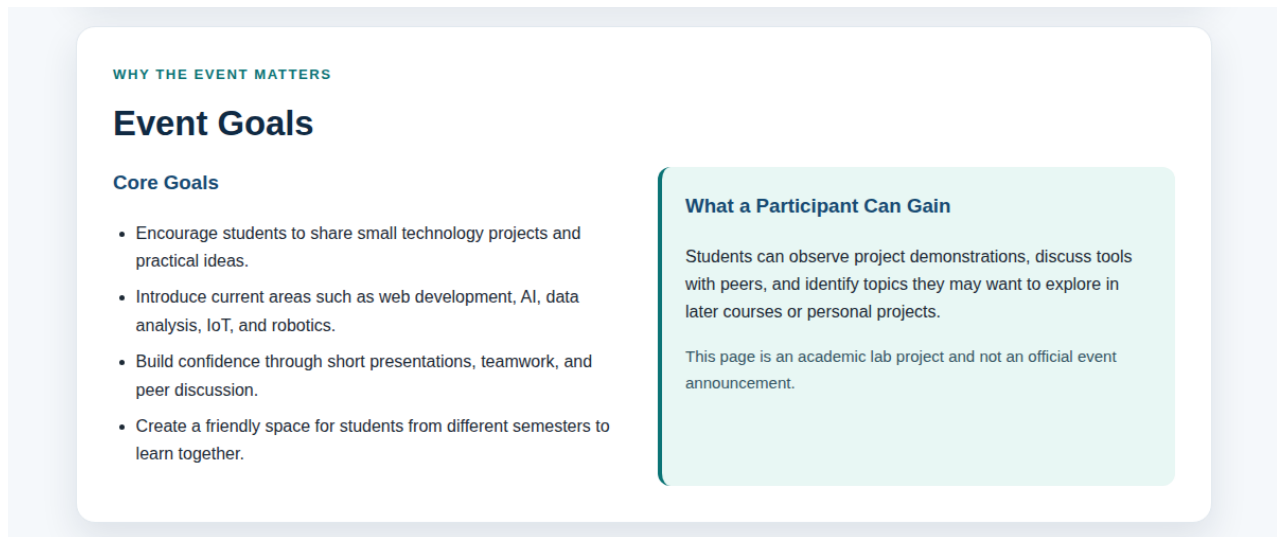


Figure 4: Unordered goals and the academic-project notice.

The image shows a mobile registration form layout. At the top, there's a navigation bar with links: "About", "Goals", "Activities", "Schedule", and a hamburger menu icon. Below the navigation bar, the form is titled "STUDENT REGISTRATION" and "Registration Form". The form contains several fields: "Full Name" with a placeholder "Enter your full name", "Student ID" with a placeholder "Enter your student ID" (highlighted with an orange focus outline), "Email Address" with a placeholder "name@example.com", "Preferred Date" with a placeholder "mm/dd/yyyy" and a calendar icon, and "Preferred Participation Type" with two radio button options: "General Attendee" and "Project Presenter".

Figure 5: Mobile registration layout with one-column fields and a visible focus outline. The navigation scrolls within its own strip.

17 Problems Faced and Solutions

Missing image asset. The initial HTML referenced an event-banner JPG that was not in the repository. An original SVG was added, and the page and documentation were updated to use its real relative path. The final browser check verifies both HTTP loading and image rendering.

A form could imply a real registration service. The previous form silently cancelled submission. The updated version states its demonstration purpose, shows local feedback and prevents data transmission. Submit is initially disabled and stays disabled when JavaScript is unavailable. The Enter-key fallback was tested separately.

Local browser access was restricted. The initial development environment blocked direct file and loopback navigation. A browser-preview run was used during development. The final suite runs in GitHub Actions against a real HTTP server, so resource loading is not inferred only from files existing on disk.

Long source listings needed readable page layout. The report includes the actual final files through listings, with wrapping and line numbers. Screenshots preserve aspect ratio. Repeated table headings keep the test cases readable across pages.

18 Learning Outcomes

The work shows how individual HTML elements combine into a complete page. Label-click and keyboard tests demonstrate the relationship between an input and its label. Goals and steps are suited to lists, while a schedule needs consistent table headings and rows. Separating HTML and CSS keeps the document structure clear while allowing responsive layouts.

The form also demonstrates an important limitation: visible input controls and browser validation do not create a registration service. A production version would require a backend, server-side validation and an appropriate data-handling process. Recording blocked checks separately from successful tests is part of reporting results accurately.

19 Conclusion

The Lab 02 page combines the required HTML5 structure, semantic sections, meaningful text, links, image, lists, schedule and registration controls. The existing design was completed without changing another lab. The actual browser results document the tested behaviour and distinguish any unavailable external check. Complete source code, five screenshots, the compiled report, reproducible build script and a verified ZIP are included in the submission.

20 References

- [1] A. Ahmad, *CSE 472 – Web and Internet Programming Lab: Complete HTML Lab Manual*, Southeast University. Supplied course manual, especially Sections 19–26. The uploaded copy was a DOCX document.
- [2] WHATWG, *HTML Living Standard*. Accessed 8 September 2026. <https://html.spec.whatwg.org/multipage/>

Image attribution: `images/event-banner.svg` is an original illustration created for this submission using simple vector shapes and text. It is not a campus photograph and contains no third-party logo or font file. The five output images are browser captures of this project.